

Kubernetes and Cloud Native Associate - KCNA

Scheduling

Node Selectors

Welcome to this guide on Kubernetes node selectors. In this article, you'll learn how to restrict pod scheduling to specific nodes based on labels. Imagine a three-node cluster where two nodes have limited hardware resources, and one node is larger and more powerful. To ensure resource-intensive tasks run on the larger node, you can utilize node selectors. By default, Kubernetes schedules pods on any available node, which might inadvertently assign a high-demand pod to a smaller node. Using node selectors, you can guarantee that pods land on nodes that meet the required label criteria.

Using Node Selectors in Pod Definitions

To restrict a pod to run only on a node labeled as large, add a `nodeSelector` field to the pod specification. Below is an example demonstrating how to define a pod with a node selector:

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
spec:
  containers:
    - name: data-processor
```

```
image: data-processor
nodeSelector:
  size: Large
```

In this example, the `nodeSelector` ensures that the pod is scheduled only on nodes having the label key `size` with the value `Large`.

Understanding Labels and Node Selectors

The key-value pair (`size: Large`) in the node selector must match the labels assigned to the nodes. Labels play a critical role in Kubernetes, not only with node selection but also for services, ReplicaSets, and deployments. It is essential to label your nodes accurately before deploying pods that use node selectors.

Labeling Your Nodes

Before creating a pod with a node selector, label the appropriate node. Use the following command to add a label to a node:

```
kubectl label nodes <node-name> <label-key>=<label-value>
```

For instance, to label a node named `node-1` as large, run:

```
kubectl label nodes node-1 size=Large
```

This command sets the label `size=Large` on `node-1`, ensuring that any pod requiring this label via its `nodeSelector` will be scheduled on the correct node.

Deploying the Pod

Once your node is labeled and your pod definition includes the node selector, deploy your pod with:

```
kubectl create -f pod-definition.yml
```

This command creates the pod named `myapp-pod`, which will be scheduled on `node-1` thanks to the matching label.



Note

Ensure that your node labels are kept up-to-date as your cluster evolves. This helps maintain consistent and predictable pod scheduling.

Limitations of Node Selectors

While node selectors are a simple and effective method for basic pod scheduling, they come with limitations. If your scheduling requirements extend beyond a single



Beta



● Search docs...



Kubernetes and Cloud Native Associate - KCNA ▾

Node Selector - Limitations



- Large OR Medium?
- NOT Small



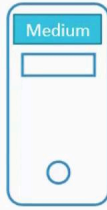
Node 1



Node 2



Node 3



Node 4



Watch Video

Watch video content

Previous

◀ Taints and Tolerations

Next

Node Affinity ▶